

2026-2 AI SOTA Paper Review Study

Vidu S2

Real-Time Interactive, Editable, and Spatial Video Generation



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4 문제로 인한 손실 현황

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6 실행 계획

1. Introduction -motivation

왜 실시간 영상 생성인가?

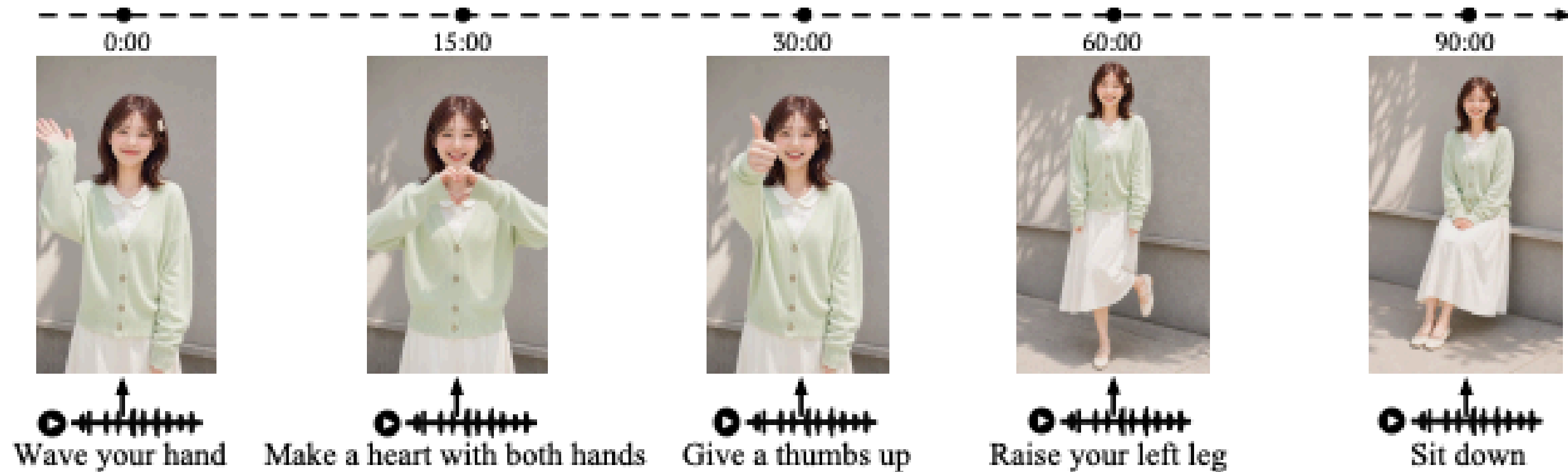


기존 영상 생성 [프롬프트 입력 → 수분간 대기 → 완성된 영상 전체 출력]

실시간 콘텐츠는 사용자마다 별도의 생성이 필요하지만, 오프라인 영상은 한 번 만든 뒤 여러 사람이 볼 수 있으므로 장기적으로는 실시간 생성 수요가 훨씬 클 것이라 주장

1. Introduction - Vidu S1

1 Speech-Guided Future Video Generation



2 Infinite Efficient Inference and Serving

540P + 42 FPS using
TurboDiffusion

Low-Cost GPU

Efficient Serving with
TurboServe

Infinite-Length
Interaction

3 Supports Various Custom Characters



1. Introduction - Vidu S1

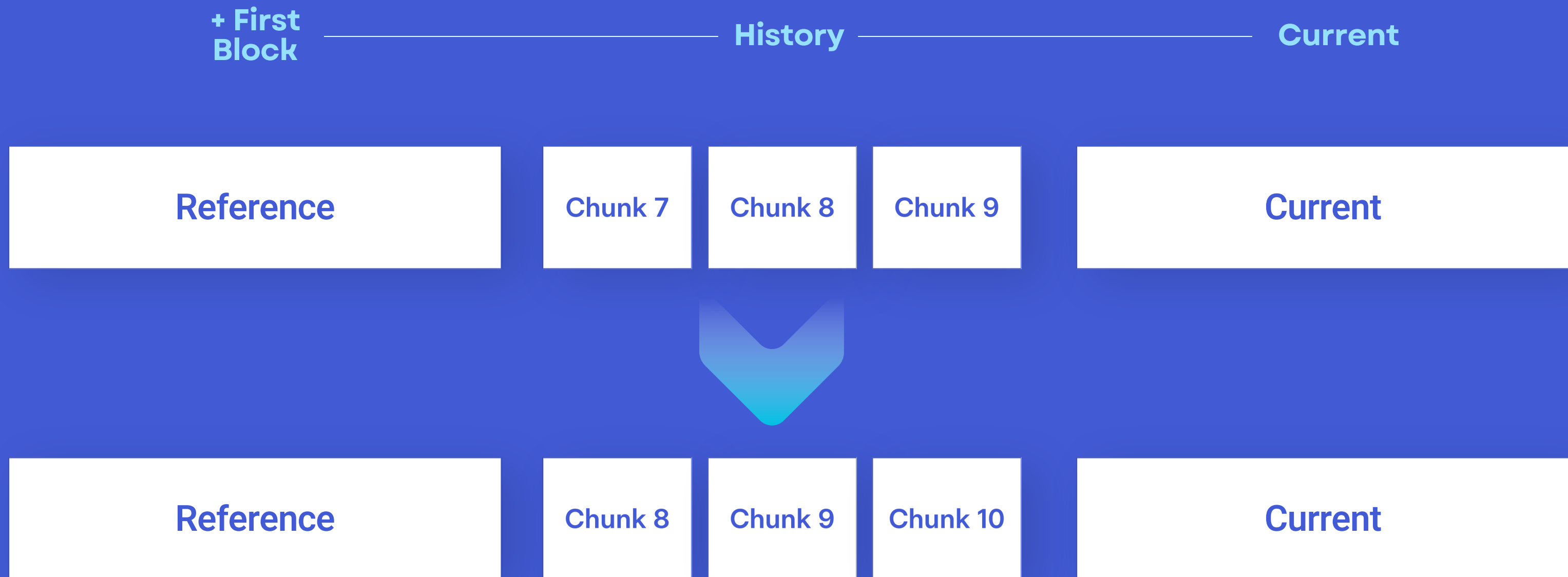
Sliding-window streaming inference



Window 크기 고정을 통한 계산량 고정

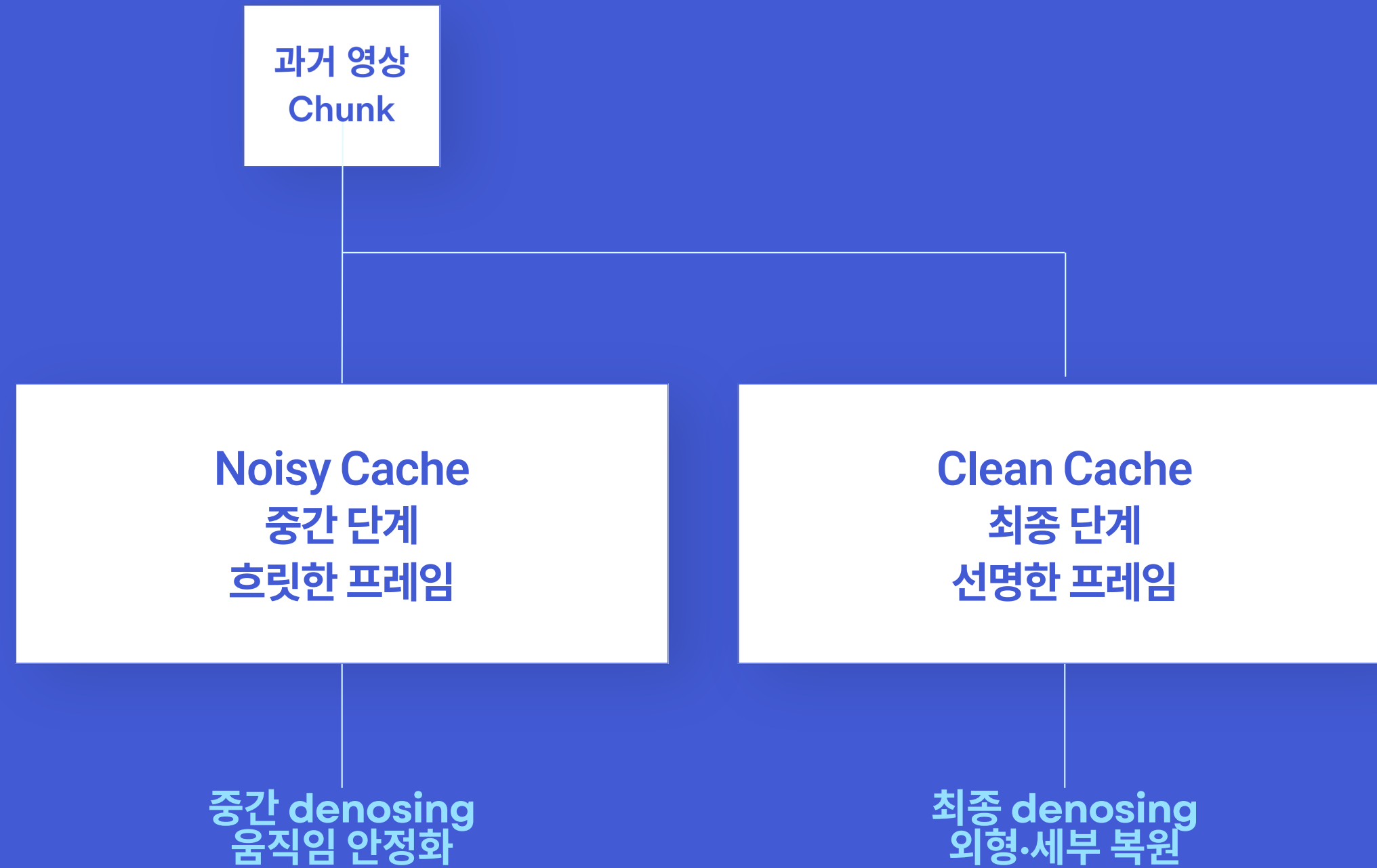
1. Introduction - Vidu S1

Persistent reference context와 RoPE Repositioning



1. Introduction - Vidu S1

TwinCache



1. Introduction - Vidu S1

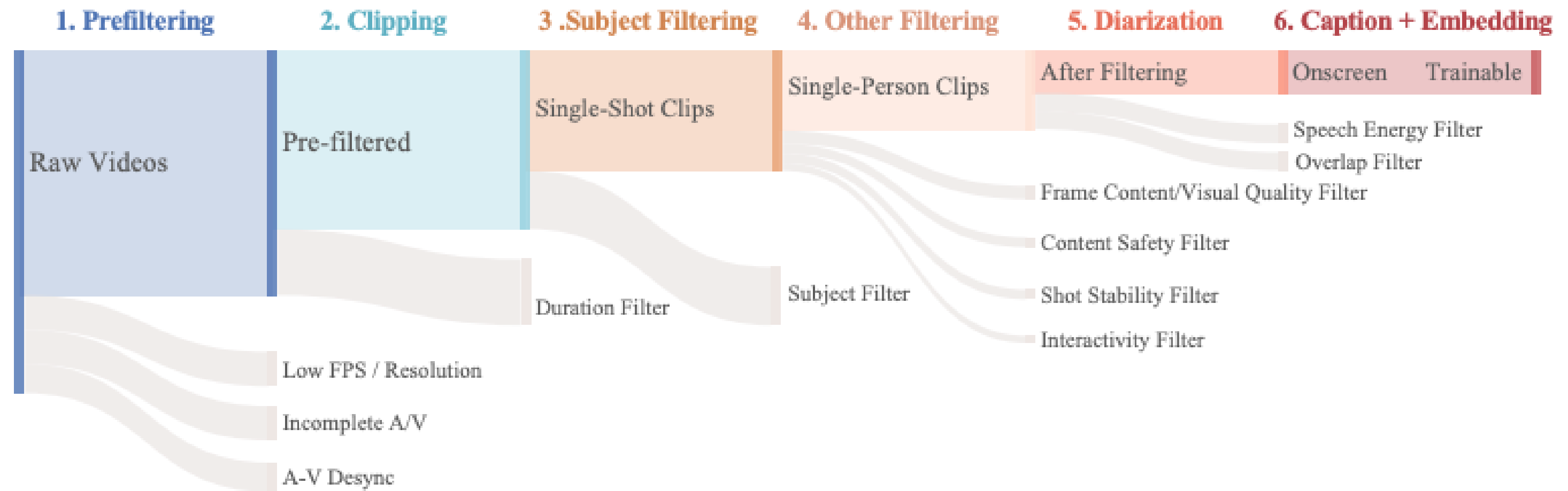
3-stage training: Bidirectional → Causal → DMD+PCM

1 **Bidirectional Teacher**

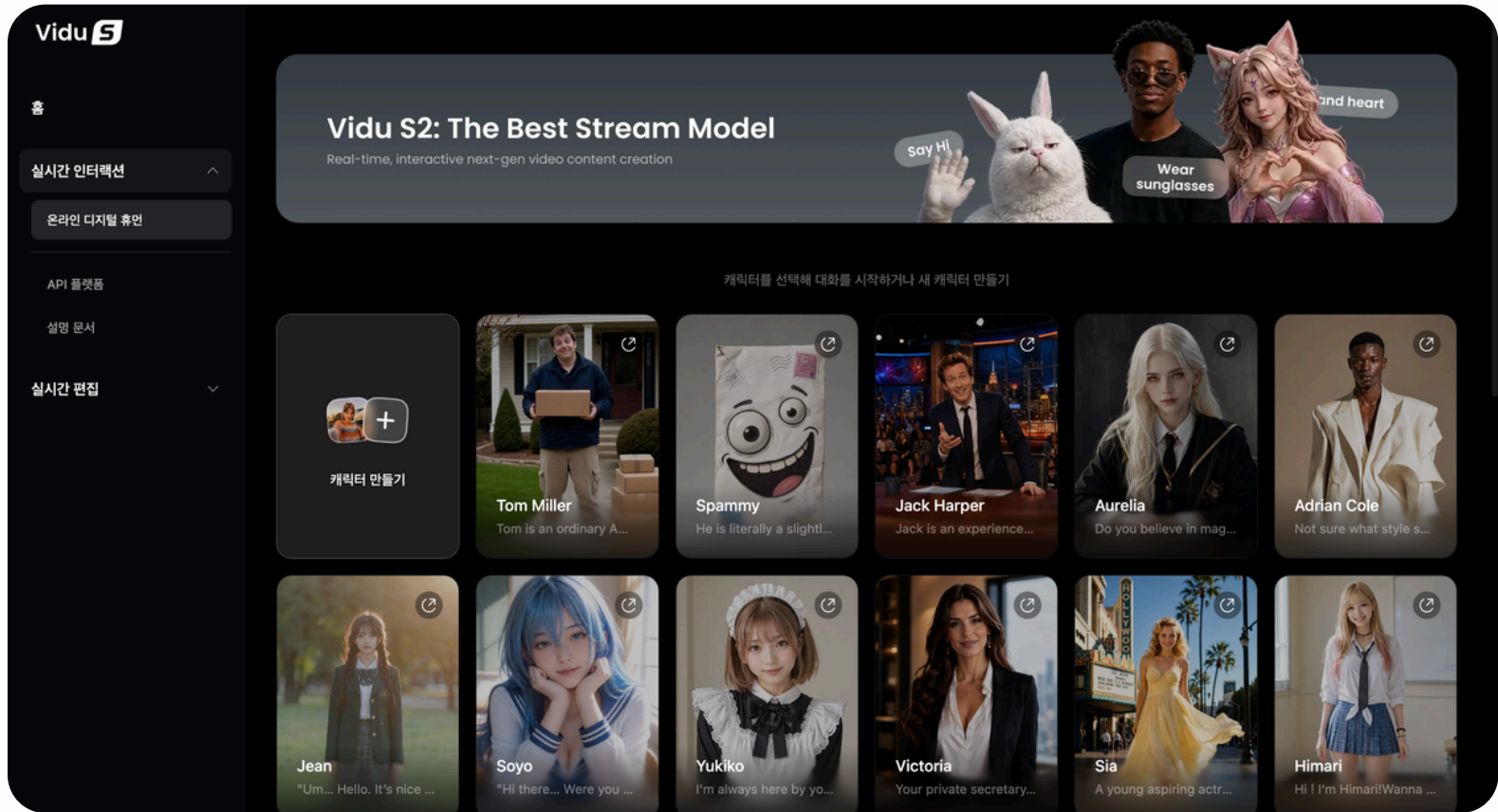
2 **Causal Teacher**

3 **DMD + PCM**

1. Introduction - Vidu S1

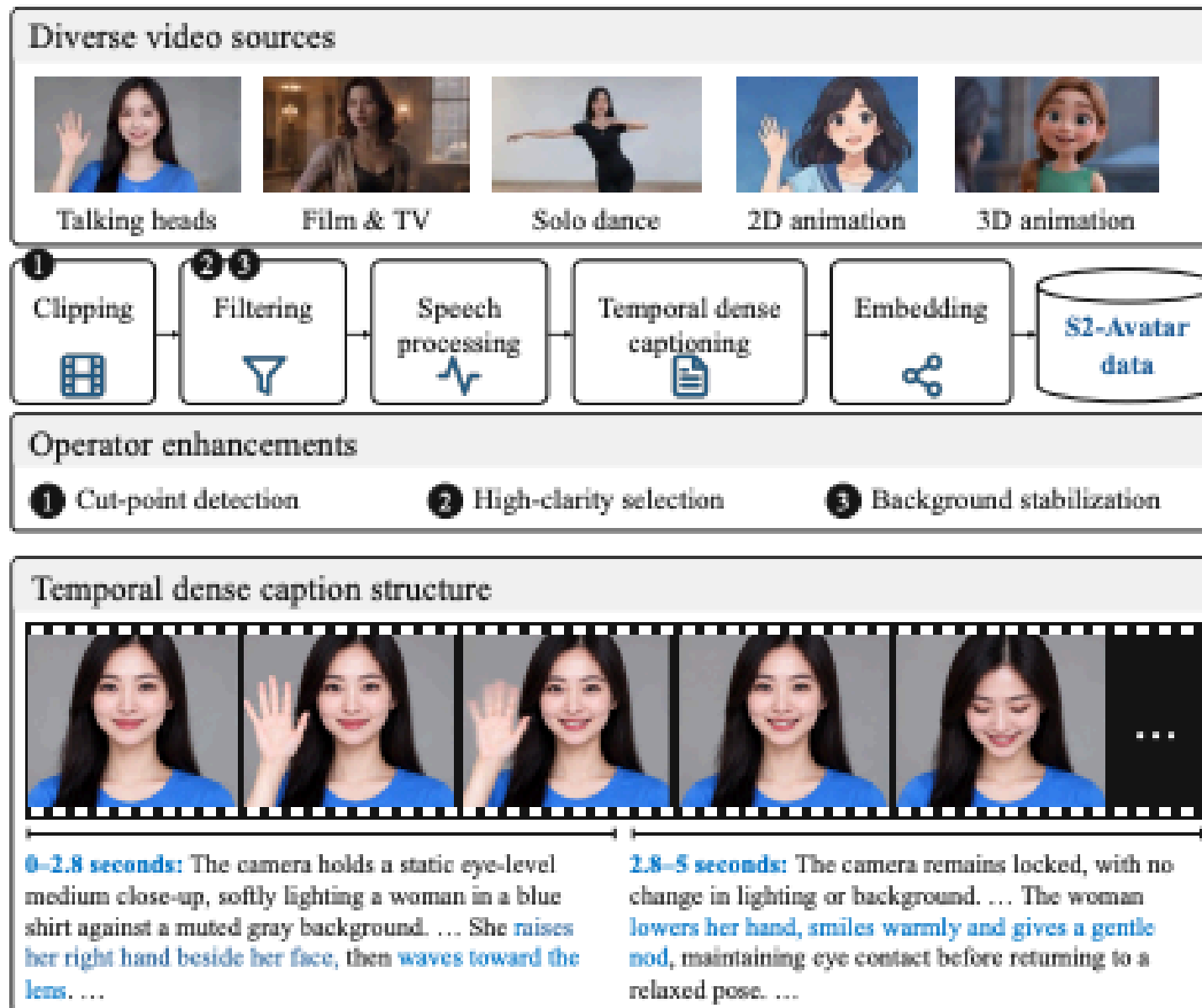


2. Vidu S2-Avatar

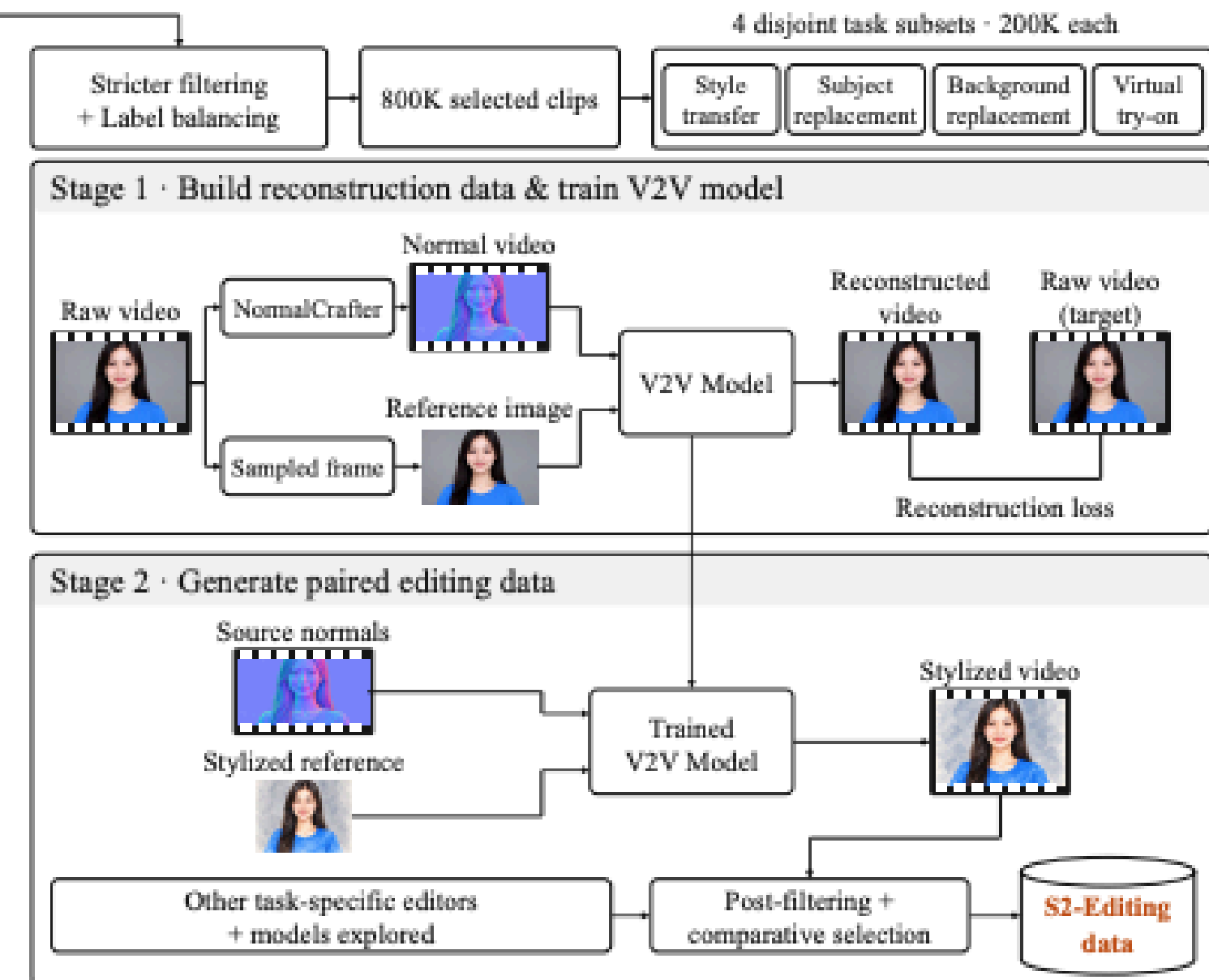


2. Vidu S2-Avatar - Data Preparation

S2-Avatar Data Pipeline



S2-Editing Data Pipeline



2. Vidu S2-Avatar - Method

- 1 Audio-Visual Joint DiT
- 2 Bidirectional I2V + R2V Training
- 3 Causal Adaptation
- 4 Self-Replay Forcing
- 5 Preference Optimization
- 6 One-step Super-Resolution Refiner

2. Vidu S2-Avatar - Method

Model Overview: Audio-Visual Joint DiT

$$\hat{x}_0^{1:N} = f_{\theta}^{\text{avatar}}(x_t^{1:N}, t, r, c^{1:N})$$

- $\backslash(r\backslash)$: 캐릭터의 참조 이미지
- $\backslash(x_i\backslash)$: $\backslash(i\backslash)$ 번째 video-audio segment
- $\backslash(c_i\backslash)$: 해당 구간의 캡션이나 사용자 지시
- $\backslash(x_t^{\{1:N\}}\backslash)$: diffusion timestep $\backslash(t\backslash)$ 에서 노이즈가 포함된 video-audio latent
- $\backslash(\hat{x}_0^{\{1:N\}}\backslash)$: 모델이 예측한 깨끗한 video-audio latent

2. Vidu S2-Avatar - Method

Bidirectional I2V + R2V Training

I2V: Image-to-Video

영상의 첫 프레임



같은 장면과 인물이
이어지는 영상 생성

R2V: Reference-to-Video

캐릭터 참조 이미지



다른 포즈와 장면에서도
같은 캐릭터 생성

2. Vidu S2-Avatar - Method

Bidirectional I2V + R2V Training

I2V: Image-to-Video

영상의 첫 프레임
↓
같은 장면과 인물이
이어지는 영상 생성

R2V: Reference-to-Video

캐릭터 참조 이미지
↓
다른 포즈와 장면에서도
같은 캐릭터 생성

$$\hat{\mathbf{x}}_0^{1:N} = \mathbf{f}_{\theta}^{\text{bi}}(\mathbf{x}_t^{1:N}, t, \mathbf{r}, \mathbf{c}^{1:N})$$

2. Vidu S2-Avatar - Method

Hybrid Teacher Forcing and Diffusion Forcing

$$\hat{x}_0^i = f_{\theta}^{\text{causal}} \left(x_{t_i}^i, t_i, r, c^i, x_{\tau_i}^{<i}, \tau_i \right)$$

Teacher Forcing

Diffusion Forcing

실제 추론에서 직접 생성한 과거 분포를
완전히 경험하지 못함

2. Vidu S2-Avatar - Method

Self-Replay Forcing

1 현재 Student가 긴 영상을 자기회귀 방식으로 생성

2 생성된 block과 KV cache를 계산 그래프에서 detach

3 생성된 영상 구간에 다시 노이즈 추가

4 여러 구간을 하나의 causal pass로 replay

$$\mathcal{L} = \mathcal{L}_{\text{DMD}}(f_{\theta}^{\text{causal}}(\mathbf{x}_t^{1:N}, t, \mathbf{r}, \mathbf{c}^{1:N})) + \mathcal{L}_{\text{perc}}(f_{\theta}^{\text{causal}}(\mathbf{x}_t^{1:N}, t, \mathbf{r}, \mathbf{c}^{1:N}))$$

5 Replay된 모든 구간에 DMD supervision 적용

6 뒤쪽 구간의 손실이 앞쪽 replay 구간까지 역전파

2. Vidu S2-Avatar - Method

Preference Optimization

Bidirectional Stage: Diffusion DPO

Streaming Stage: Streaming NFT

2. Vidu S2-Avatar - Method

One-Step Super-Resolution Refiner

$$\mathbf{z}_{t_r}^i = (1 - t_r)\mathcal{U}(\hat{\mathbf{x}}_{0,\text{LR}}^i) + t_r\boldsymbol{\epsilon}^i, \quad \boldsymbol{\epsilon}^i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

2. Vidu S2-Avatar - Method

Inference Infrastructure

1 **Efficient Attention**

2 **W8A8 Linear Layers**

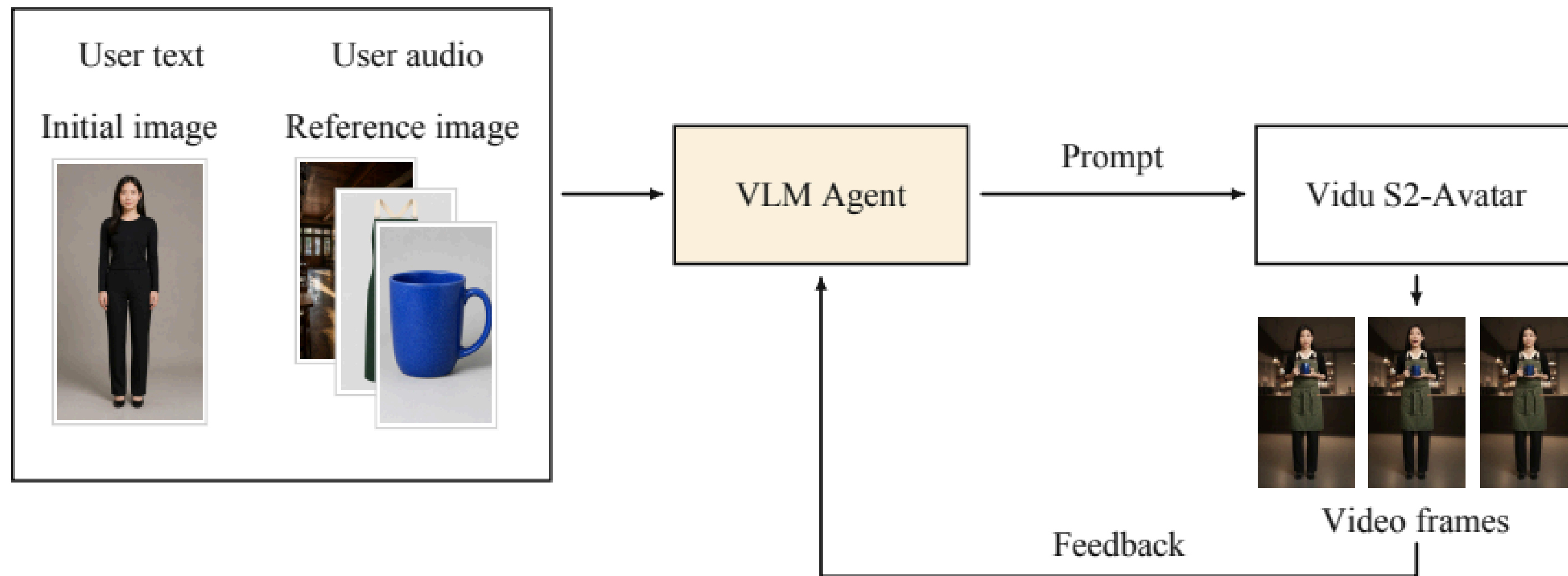
3 **Kernel Fusion**

4 **CUDA Graphs**

5 **Multi-GPU Parallelism**

2. Vidu S2-Avatar - Method

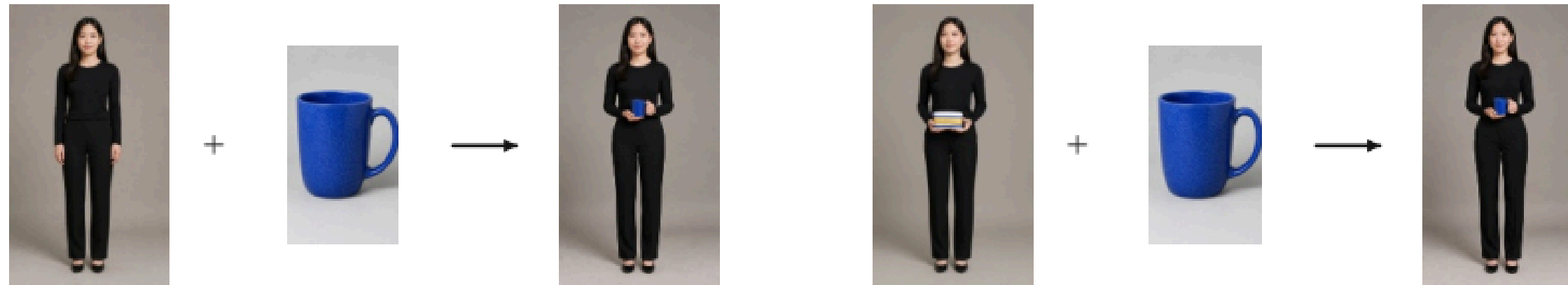
gentic System



2. Vidu S2-Avatar - Method

gentic System

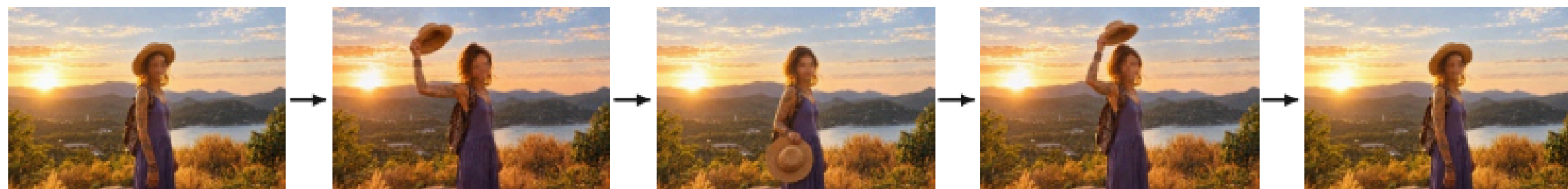
a. Reference Object Generation & Replacement



b. Reference Scene Transition



c. Taking Off and Putting On Accessories



3. Vidu S2-Editing

전체 구조



Vidu S2-Editing

API Docs

Vidu S2-Editing

Enables real-time style, outfit, subject, and background editing of input video streams based on style, outfit, subject, and background reference images.

Try Now

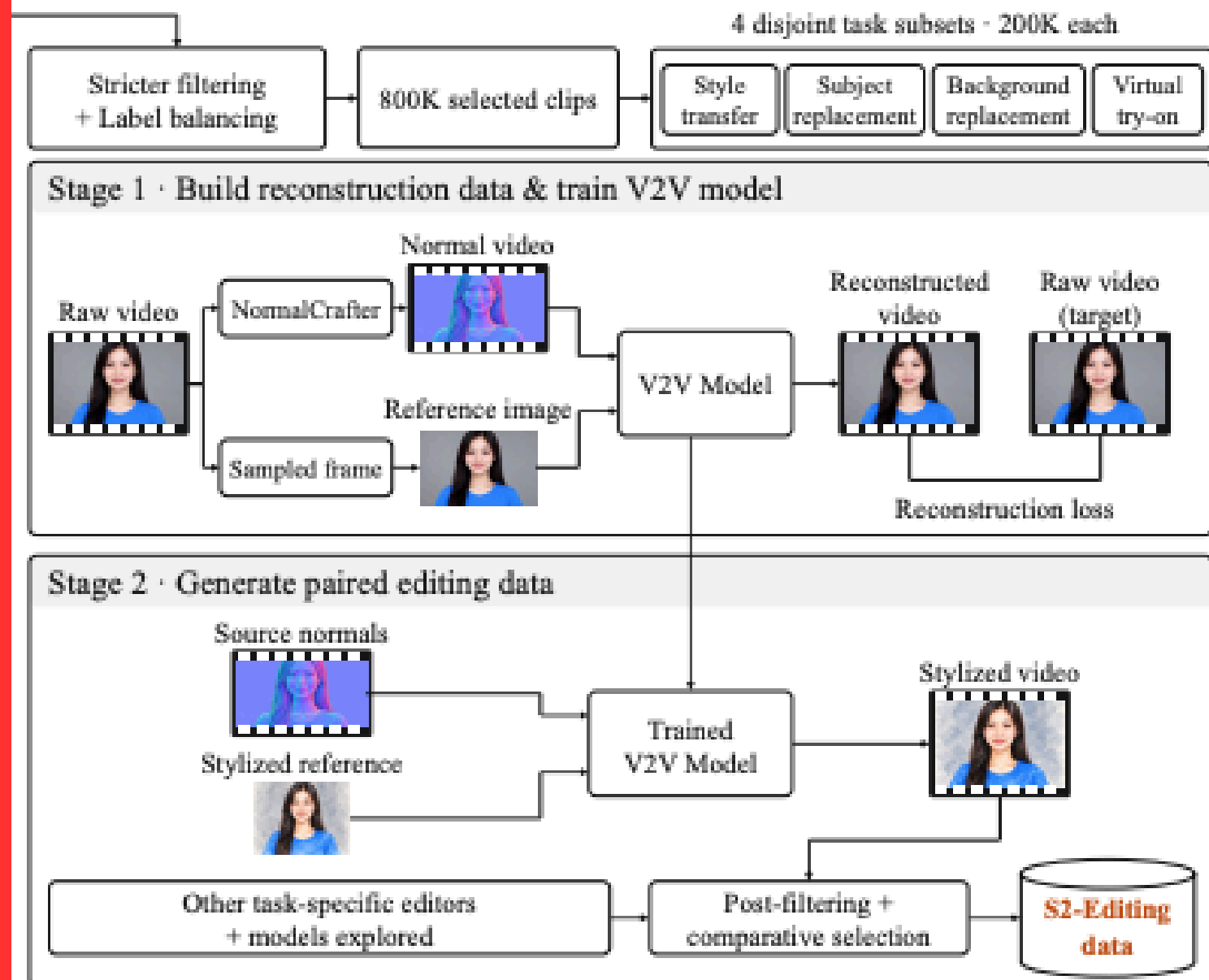
3. Vidu S2-Editing

전체 구조

S2-Avatar Data Pipeline



S2-Editing Data Pipeline



3. Vidu S2-Editing - Method

Model Overview

$$\hat{\boldsymbol{x}}_0^{1:N} = \boldsymbol{f}_{\boldsymbol{\theta}}^{\text{edit}} \left(\boldsymbol{x}_t^{1:N}, t, \boldsymbol{s}^{1:N}, \boldsymbol{r}, \boldsymbol{c}^{1:N} \right)$$

3. Vidu S2-Editing - Method

Model Overview

Bidirectional Video Editing Training

Causal Streaming Training

4. Real-Time Spatial Video Generation and Editing

Model Overview

Spatial Video Example (Vidu S2-Avatar)



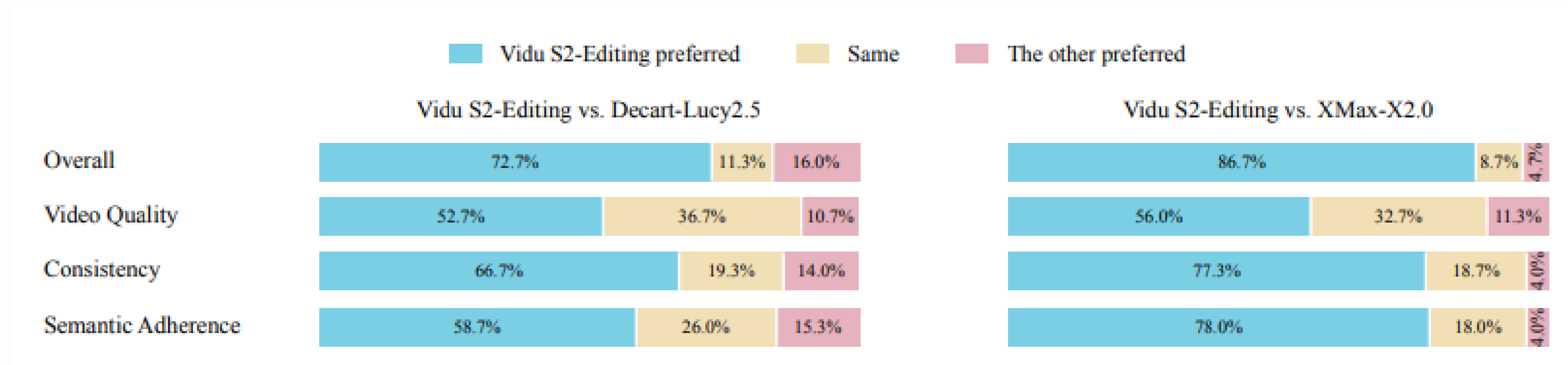
Spatial Video Example (Vidu S2-Editing)



5. Experiments

Model	VA↑	VQ↑	PQ↑	AQ↑	AVAlign↑	AVSync↓	AIF↑	SC↑	BC↑
SoulX-FlashTalk	.281	2.515	7.091	3.006	.121	<u>.648</u>	2.499	.946	.964
AptAvatar	.458	2.040	7.075	2.582	.099	1.284	2.448	.860	.923
OmniForcing	.554	2.374	6.351	2.528	.119	1.423	2.343	.962	.953
Hallo-Live	.574	–	5.396	–	.102	1.353	–	.991	.977
Self-Forcing	.585	2.753	6.453	2.623	.256	.919	2.810	.981	.969
IAMFlow	.602	2.840	6.415	2.625	.268	1.016	2.789	.986	.973
LongLive	.605	2.768	6.459	2.696	<u>.272</u>	.969	2.814	.986	.973
SWIFT	.605	2.735	6.483	2.680	.270	.970	<u>2.887</u>	.986	.973
AvatarForcing	.639	–	7.070	–	.089	1.041	–	.987	.976
PixVerse R1	.529	2.428	6.348	2.810	.234	.855	2.511	.907	.930
Odyssey-2	.531	2.820	6.426	2.659	.260	.946	2.778	.976	.966
HappyOyster	.529	2.785	6.817	<u>3.157</u>	.206	1.044	2.692	.904	.934
Live Avatar	<u>.661</u>	<u>3.295</u>	<u>7.133</u>	3.079	.116	1.145	2.745	<u>.997</u>	<u>.989</u>
Vidu S2-Avatar	.687	3.370	7.138	3.286	.353	.617	2.985	.998	.993

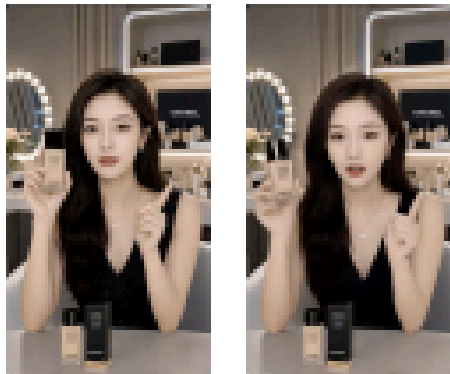
5. Experiments



5. Experiments

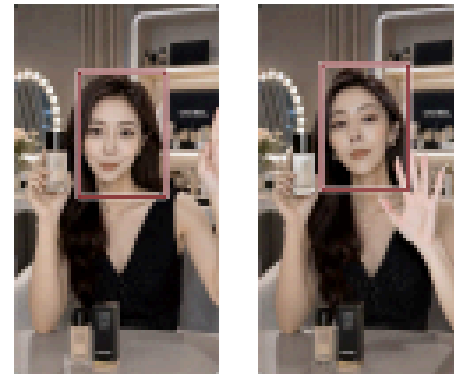
Temporal consistency

Vidu S2-Avatar



Good: Stable identity, hair, attire, and scene details.

PixVerse Image Avatar



Bad: Facial identity and body proportions drift.

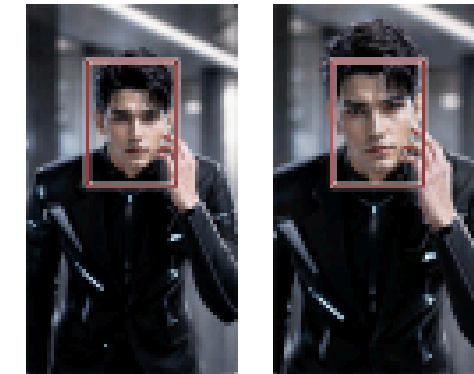
Temporal consistency

Vidu S2-Avatar



Good: Stable identity, hair, eye color, and attire.

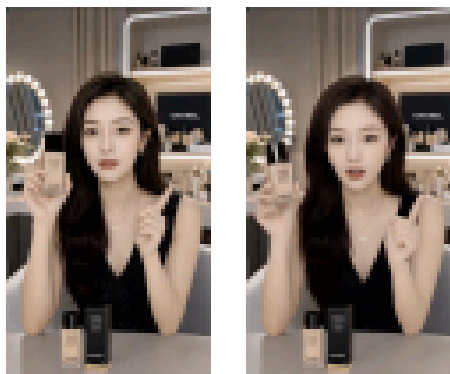
PixVerse Image Avatar



Bad: Face, framing, hairstyle, and attire drift.

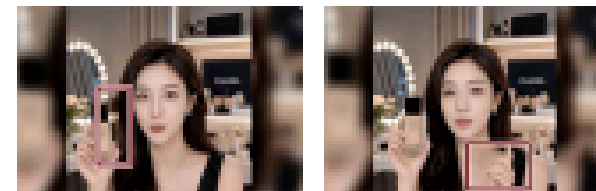
Visual stability

Vidu S2-Avatar



Good: Stable product geometry and hand structure.

Runway Character GWM-1



Bad: Hair and fingers are distorted.

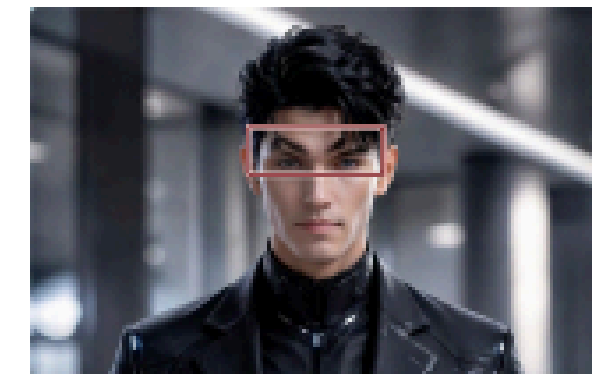
Visual stability

Vidu S2-Avatar



Good: Stable eyes and facial structure.

Runway Character GWM-1



Bad: Distorted eyebrows.

6. Conclusion

Vidu S2-Avatar
Vidu S2-Editing



감사합니다



궁금한 점이 있으시면 언제든지 연락주세요.

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